ChipGuard :-

Q2 A, B, C ⇒ PU

Inputs
- CLK
- RESET
- START
- A
- B
- C
- READY

Transmitter

Output
- DATA_OUT [7:0] } ⟶ Reciever
- VALID
- DONE } ⟶ transmitter

READY { 1 : Accepts Data
0 : Busy

Reciever

States :-

IDLE :- VALID = 0 DONE = 0, RESET = 1
condition

TRANSFER :- START = 1 ⇒ VALID = 1
(Astate) Condition set

↳ If READY = 0, no change.
VALID = 1

If READY = 1,
VALID = 1, DATA_OUT = A,
change state

TRANSFER
(Bstate) :- If READY = 0, No Change :-
VALID = 1

If READY = 1,

VALID = 1, DATA OUT = B

change state

TRANSFER :

(C state)

If READY = 0, no change

VALID = 1

If READY = 1,

VALID = 1, DATA_OUT = C,

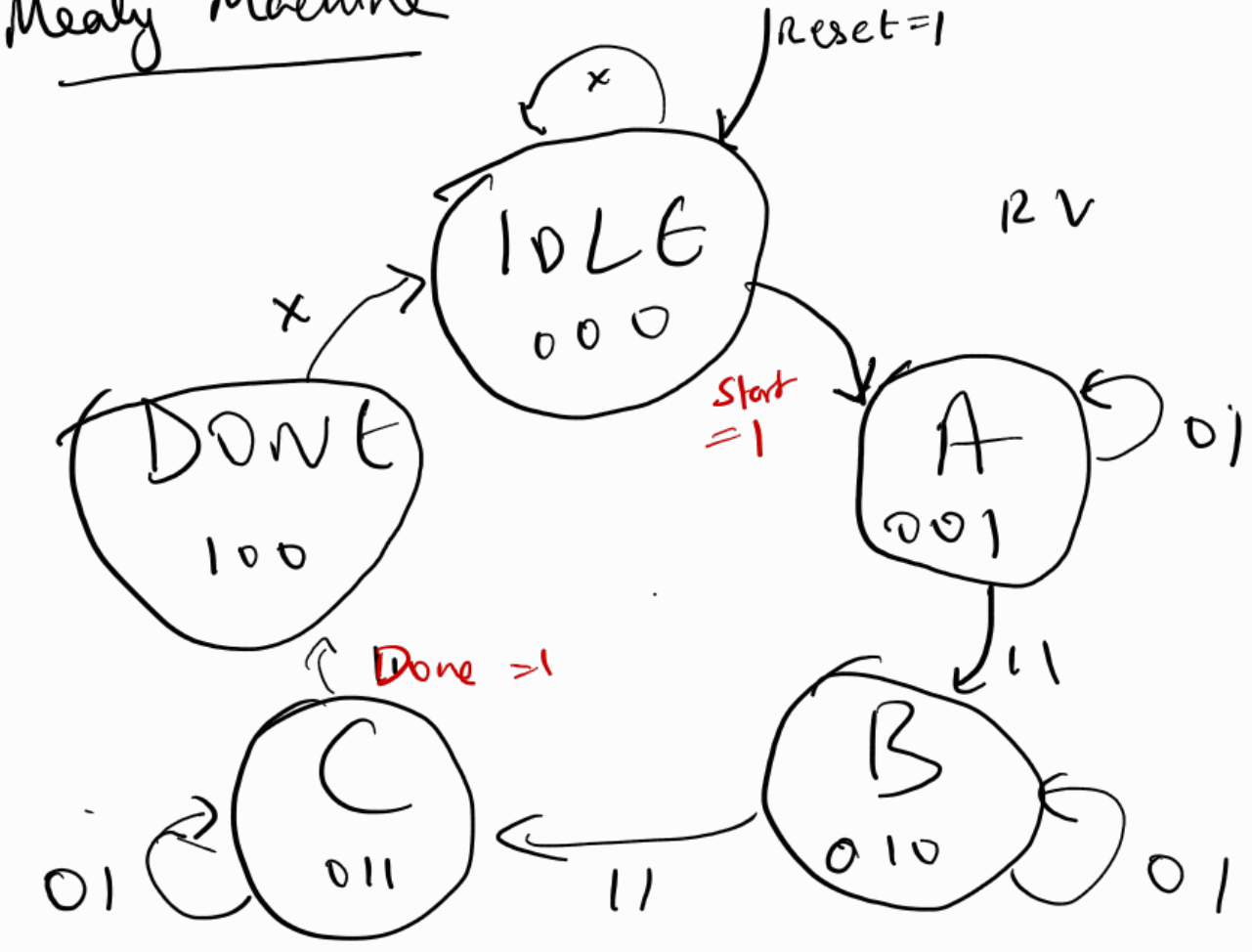
DONE = 1

changes state

DONE = VALID = 0

change state

Mealy Machine



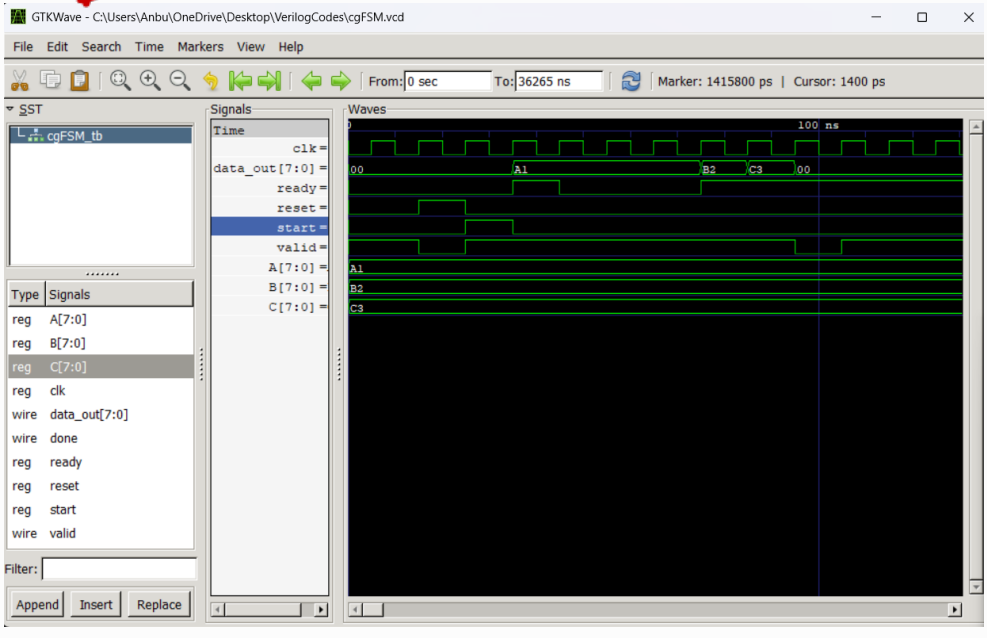
# State Transition Table :-

| CS | $I_1$ $I_2$ | N.S | O |
|---|---|---|---|
| IDLE | X 1 | A | — |
| A | 0 X | A | A |
| A | 1 X | B | A |
| B | 0 X | B | B |
| B | 1 X | C | B |
| C | 0 X | C | C |
| C | 1 X | DONE | C |
| DONE | X X | IDLE | — |

$I_1$ Ready

$I_2$ Start

Valid assumed to be 1 throughout



7)

| Rising Edge | Ready | Reading on DATA_OUT | Transfer? |
|---|---|---|---|
| 1 | 0 | A | No |
| 2 | 1 | A | Yes |
| 3 | 0 | B | No |
| 4 | 0 | B | No |
| 5 | 1 | B | Yes |
| 6 | 1 | C | Yes |